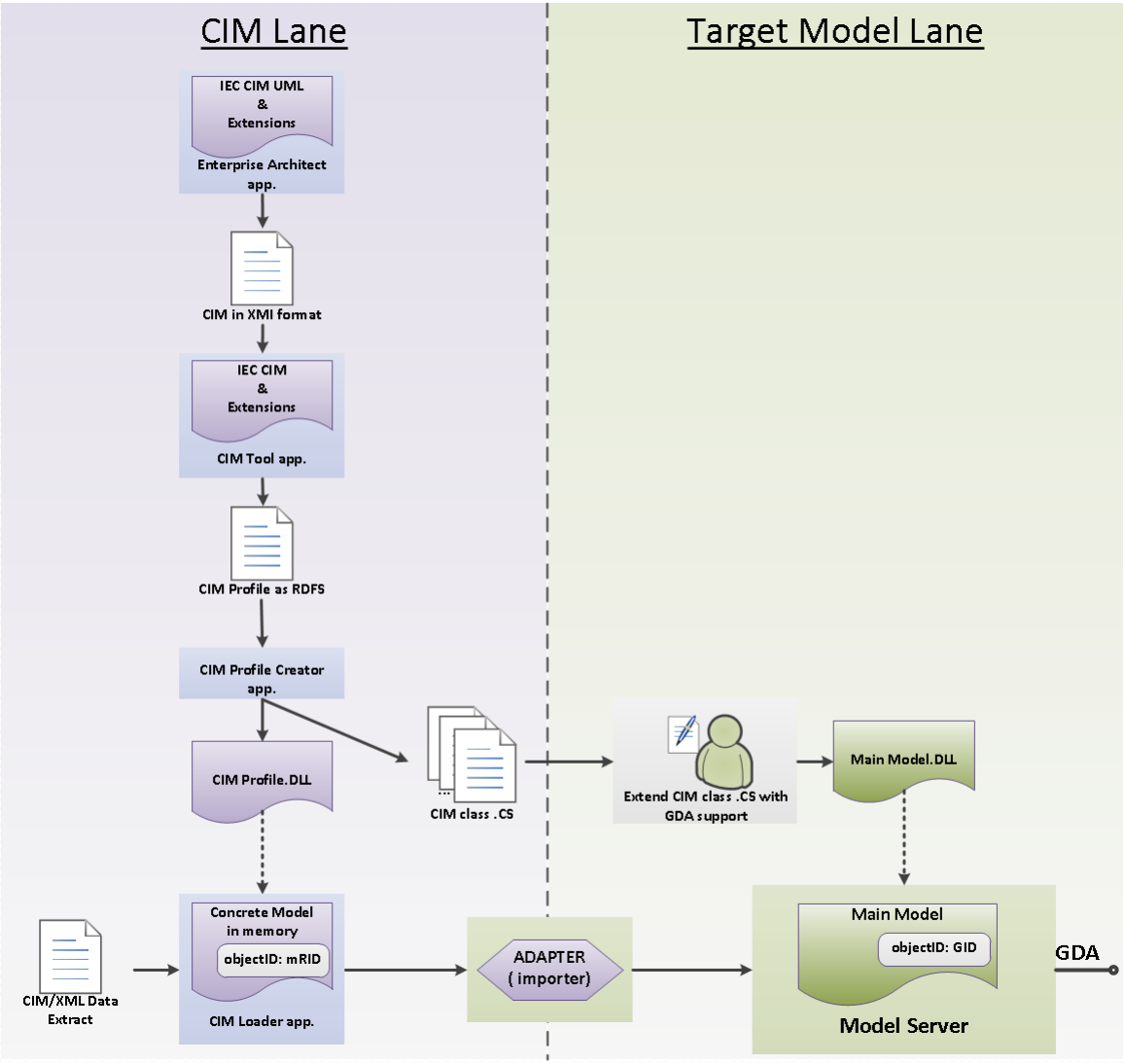
Modeli podataka u pametnim mrežama

Vežba 5:
ModelCode,
Network Model servis

Tok podataka pri inicijalizaciji modela elektroenergetske mreže



Kreiranje Network Model servisa

- Model zasnovan na CIM-u
- Identifikacija objekata
- Referenciranje objekata

Implicitno definisanje konkretnih klasa 1/2

- *DMSType* enumeracija

```
public enum DMSType : short
{
    BASEVOLTAGE                                 = 0x0001,
    LOCATION                                    = 0x0002,
    POWERTR                                     = 0x0003,
    POWERTRWINDING                              = 0x0004,
    WINDINGTEST                                 = 0x0005,
}
```

- 16-bitna enumeracija
- Svaka klasa čije se instanciranje očekuje (**konkretna klasa**) dobija odgovarajuću vrednost u *DMSType* enumeraciji

Implicitno definisanje konkretnih klasa 2/2

- … *DMSType* enumeracija
  - *BaseVoltage* - *DMSType.BASEVOLTAGE*
  - *Location* - *DMSType.LOCATION*
  - *PowerTransformer* - *DMSType.POWERTR*
  - *TransformerWinding* - *DMSType.POWERTRWINDING*
  - *WindingTest* - *DMSType.WINDINGTEST*

Implicitno definisanje hijararhije nasleđivanja1/2

- …*ModelCode* enumeracija
  - 64-bitna enumeracija

Nasledjivanje	DMSType	Opis atributa
32 bita	16 bita	16 bita

  - Svaki tip resursa u modelu se jednoznačno identifikuje odgovarakućim *ModelCode*-om:
    - Svaka klasa(apstraktna ili konkretna) dobija svoj *ModelCode*

      PowerSystemResource - *ModelCode.PSR*

      PowerTransformer - *ModelCode.POWERTR*
    - Svaki atribut (svake klase) dobija svoj *ModelCode*

      PowerTransformer.Function - *ModelCode.POWERTR_FUNC*

      PowerSystemResource .CustomType - *ModelCode.PSR_CUSTOMTYPE*

Implicitno definisanje hijararhije nasleđivanja2/2

- *ModelCode* enumeracija

```
public enum ModelCode : long
{
    IDOBJ                                   = 0x1000000000000000,
    IDOBJ_GID                               = 0x1000000000000104,
    IDOBJ_DESCRIPTION                       = 0x1000000000000207,
    IDOBJ_MRID                              = 0x1000000000000307,
    IDOBJ_NAME                              = 0x1000000000000407,

    PSR                                     = 0x1100000000000000,
    PSR_CUSTOMTYPE                          = 0x1100000000000107,
    PSR_LOCATION                            = 0x1100000000000209,

    ...

    POWERTRWINDING_PHASETOGRNDVOLTAGE       = 0x1111000000040605,
    POWERTRWINDING_PHASETOPHASEVOLTAGE      = 0x1111000000040705,
    POWERTRWINDING_POWERTRW                 = 0x1111000000040809,
    POWERTRWINDING_TESTS                    = 0x1111000000040919,
}
```

ModelCode semantika 1/8

- Vrednost ModelCode-a nosi u sebi niz informacija vezanih za resurs
- *ModelCode* enumeracija – nasleđivanje
  - Najviših 32 bita opisuje nasleđivanje.
  - "*Child*" klasa **UVEK** ima jednu cifru više od "*parent*" klase
  - Ukoliko više klasa nasleđuje istog *parent*-a, dodatna cifra se uveća za svaku "*child*" klasu.

ModelCode.IDOBJ = 0x**10000000**00000000;

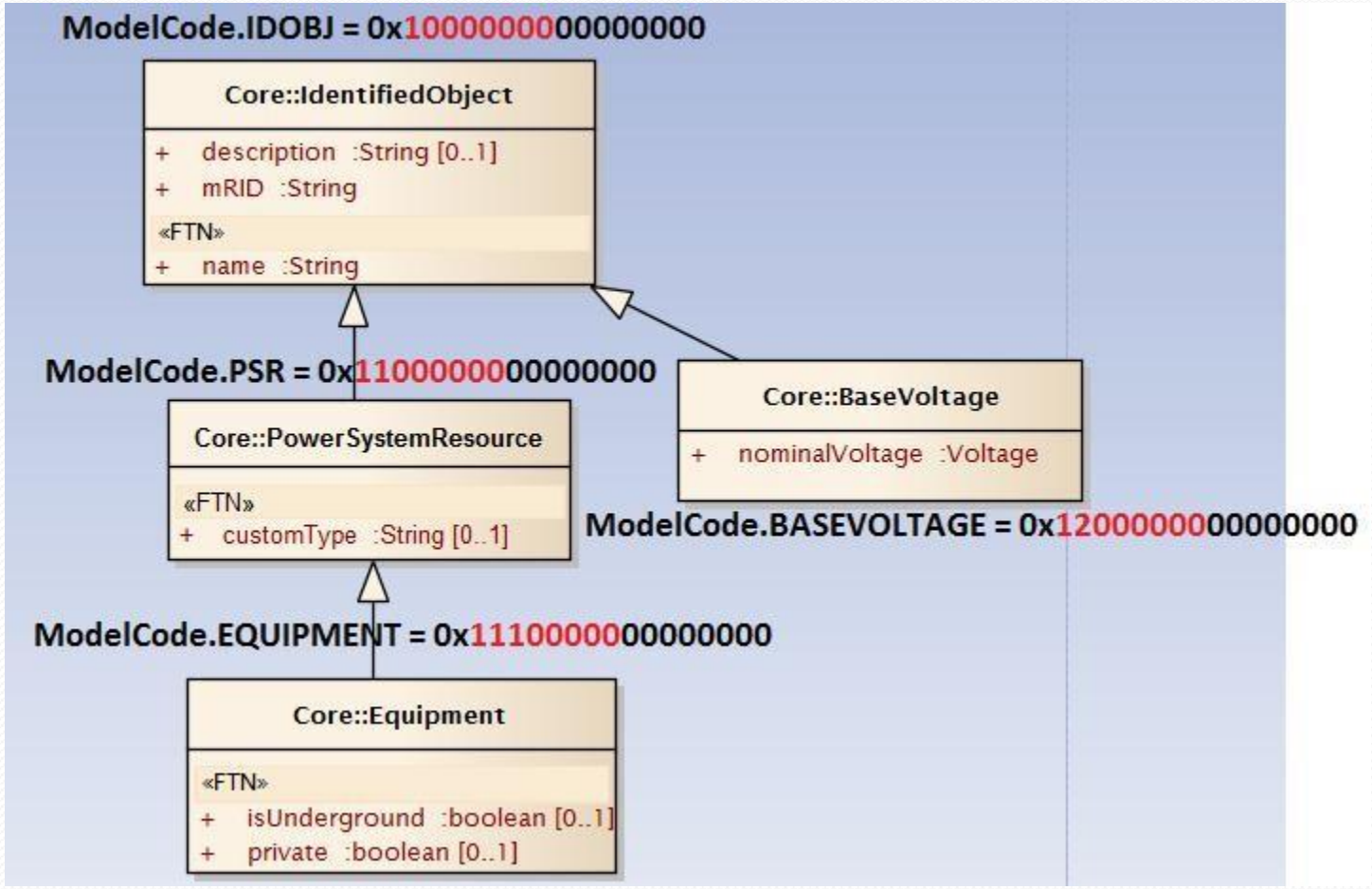
ModelCode.PSR = 0x**11000000**00000000;

ModelCode.BASEVOLTAGE = 0x**12000000**00010000;

ModelCode.EQUIPMENT = 0x**11100000**00000000;

ModelCode semantika 2/8

- …*ModelCode* enumeracija – nasleđivanje



ModelCode semantika 3/8

- *ModelCode* enumeracija – apstraktni i konkretni tipovi
  - Narednih 16 bita opisuje da li je klasa apstraktna ili ne.
  - Ukoliko je klasa apstraktna vrednost je postavljena na nulu, u suprotnom vrednost odgovara vrednosti *DMType*-a za tu konkretnu klasu.

ModelCode.IDOBJ	= 0x10000000**0000**0000;
ModelCode.PSR	= 0x11000000**0000**0000;
ModelCode.BASEVOLTAGE	= 0x12000000**0001**0000;
DMSType.BASEVOLTAGE	= 0x**0001**;
ModelCode. LOCATION	= 0x12000000**0002**0000;
DMSType. LOCATION	= 0x**0002**;

ModelCode semantika 4/8

- *ModelCode* enumeracija –atributi
  - Svaki atribut neke klase (apstraktne ili konkretne) ima svoj *ModelCode*

    IdentifiedObject.Description - *ModelCode.IDOBJ_DESCRIPTION*

    BaseVoltage.NominalVoltage - *ModelCode.BASEVOLTAGE_NOMINALVOLTAGE*
  - Veći deo vrednosti *ModelCode*-a koji je dodeljen atributu (viših 48 bita) je identično *ModelCode*-u koji je dodeljen klasi kojoj atribut pripada

    ModelCode.IDOBJ_DESCRIPTION = 0x**100000000000**0207

    ModelCode.IDOBJ = 0x**100000000000**0000
  - Najnižih 16 bita rezervisani su za opis atributa

    ModelCode.IDOBJ_DESCRIPTION = 0x100000000000**0207**

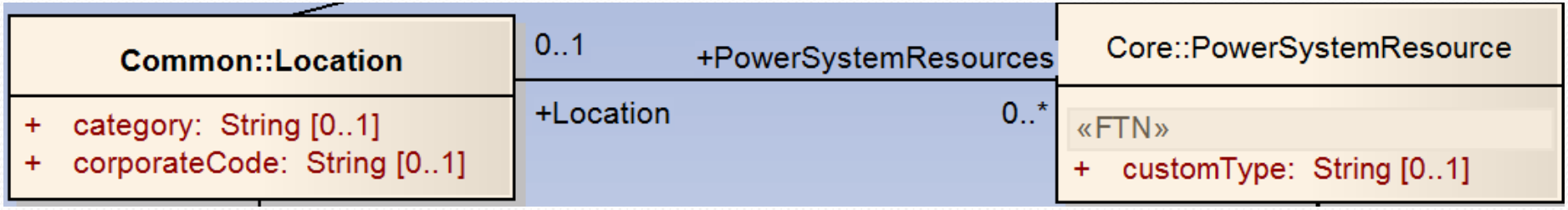
    ModelCode.BASEVOLTAGE_NOMINALVOLTAGE = 0x100000000001**0105**
  - Ukoliko je *ModelCode* dodeljen klasi najnižih 16 bita su uvek nula

    ModelCode.IDOBJ = 0x100000000000**0000**

    ModelCode.BASEVOLTAGE = 0x120000000001**0000**

ModelCode semantika 5/8

- ...*ModelCode* enumeracija – atributi
  - Viših 8 bita označavaju redni broj atributa u klasi



ModelCode.PSR = 0x1100000000000000

ModelCode.PSR_CUSTOMTYPE = 0x1100000000000107

ModelCode.PSR_LOCATION = 0x1100000000000209

...

ModelCode.LOCATION = 0x1300000000020000

ModelCode.LOCATION_COORPORATECODE = 0x1300000000020107

ModelCode.LOCATION_CATEGORY = 0x1300000000020207

ModelCode.LOCATION_PSRS = 0x1300000000020319

ModelCode semantika 6/8

- ...*ModelCode* enumeracija –atributi
  - Nižih 8 bita označavaju tip atributa u klasi
  - Ukoliko je u pitanju prost tip:
    - prva vrednost je nula,
    - druga vrednost označava koji je prost tip u pitanju (*bool, int, long, string*)
  - Ukoliko je u pitanju lista prostih tipova:
    - prva cifra je uvek jedinica,
    - druga označava koje proste vrednosti sadrži lista.

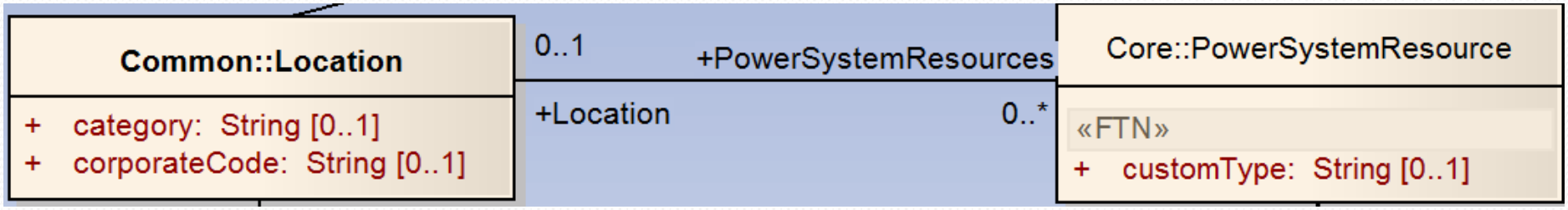
```
public enum PropertyType : short
{
    Empty = 0,

    Bool        = 0x01,
    Byte        = 0x02,
    Int32       = 0x03,
    Int64       = 0x04,
    Float       = 0x05,
    Double      = 0x06,
    String      = 0x07,
    DateTime    = 0x08,
    Reference   = 0x09,
    Enum        = 0x0a,
    TimeSpan    = 0x0c,

    BoolVector      = 0x11,
    ByteVector      = 0x12,
    Int32Vector     = 0x13,
    Int64Vector     = 0x14,
    FloatVector     = 0x15,
    DoubleVector    = 0x16,
    StringVector    = 0x17,
    DateTimeVector  = 0x18,
    ReferenceVector = 0x19,
    EnumVector      = 0x1a,
    TimeSpanVector  = 0x1c,
}
```

ModelCode semantika 7/8

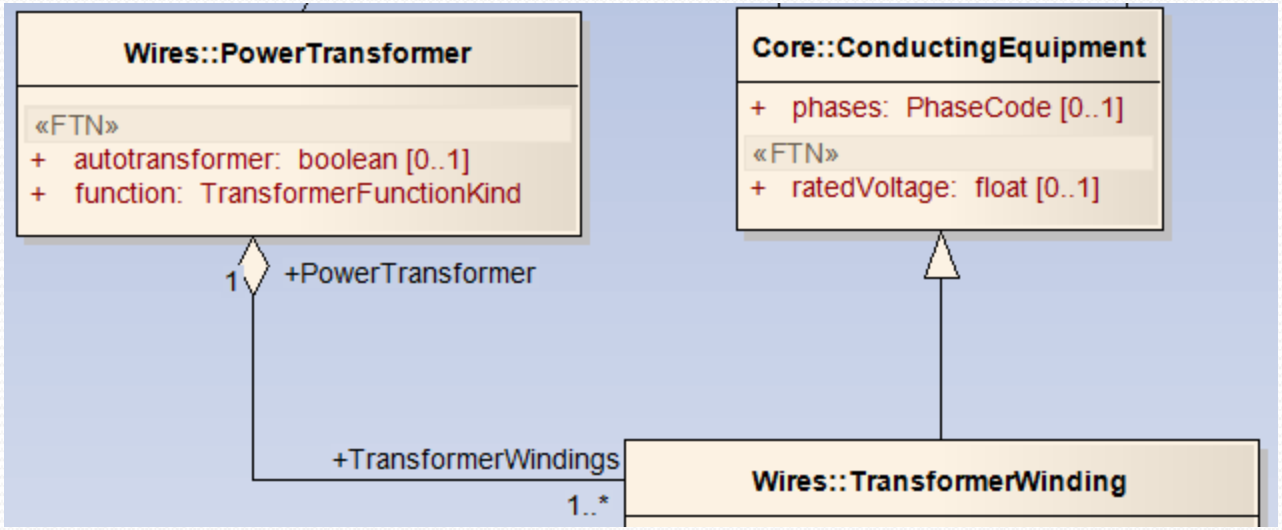
- ...*ModelCode* enumeracija –atributi
  - Nižih 8 bita označavaju tip atributa u klasi (int, long, string, List<int>...)



ModelCode.PSR = 0x1100000000000000
ModelCode.PSR_CUSTOMTYPE = 0x1100000000000107
ModelCode.PSR_LOCATION = 0x1100000000000209
...
ModelCode.LOCATION = 0x1300000000020000
ModelCode.LOCATION_COORPOATECODE = 0x1300000000020107
ModelCode.LOCATION_CATEGORY = 0x1300000000020207
ModelCode.LOCATION_PSRS = 0x1300000000020319

ModelCode semantika 8/8

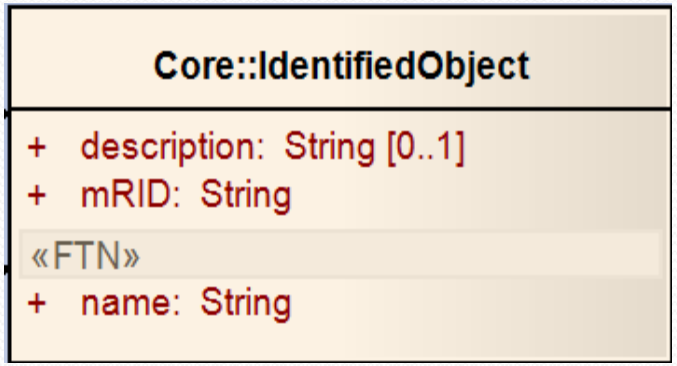
- …*ModelCode* enumeracija – atributi



ModelCode. POWERTR	= 0x1112000000030000
ModelCode. POWERTR_FUNC	= 0x111200000003010a
ModelCode. POWERTR_AUTO	= 0x1112000000030201
ModelCode. POWERTR_WINDINGS	= 0x1112000000030319

Globalni identifikator (GID) 1/2

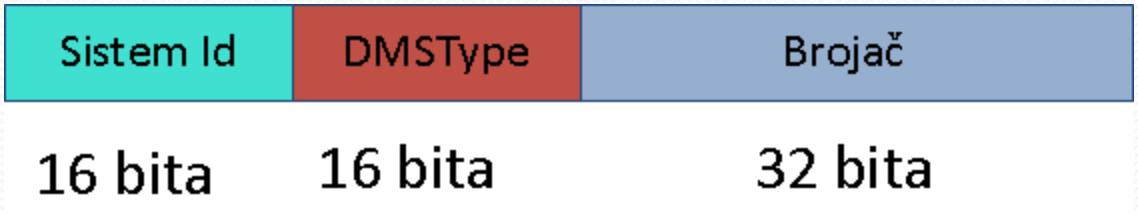
- Globalni identifikator
  - Iako je mRID jedinstven za svaki entitet koji se kreira, njegova upotreba za identifikaciju entiteta može da uspori rad servisa jer je u pitanju string
  - Uvodi se novi jedinstveni generator tipa long (poređenje brojeva je daleko brža od poređenja stringova)
  - Globalni identifikator je definisan u klasi *IdentifiedObject* koja se nalazi u hijerarhiji nasleđivanja svih ostalih klasa. Iz tog razloga svaki entitet sadrži globalni identifikator



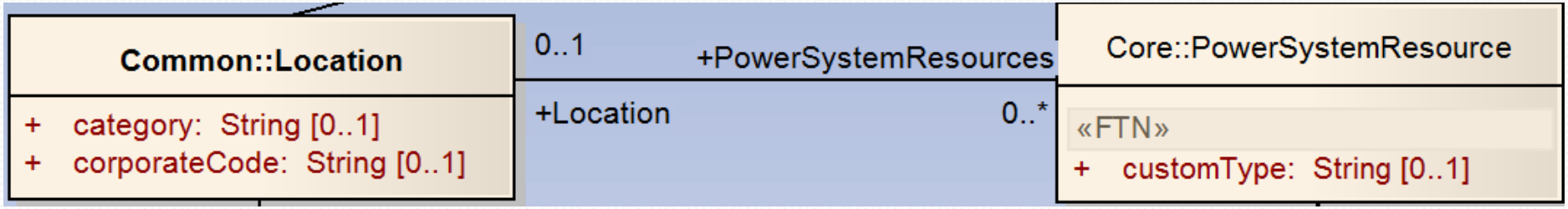
ModelCode.IDOBJ	= 0x1000000000000000
ModelCode. IDOBJ_GID	**= 0x1000000000000104**
ModelCode. IDOBJ_DESCRIPTION	= 0x1000000000000207
ModelCode. IDOBJ_MRID	= 0x1000000000000307
ModelCode. IDOBJ_NAME	= 0x1000000000000407

Globalni identifikator (GID) 2/2

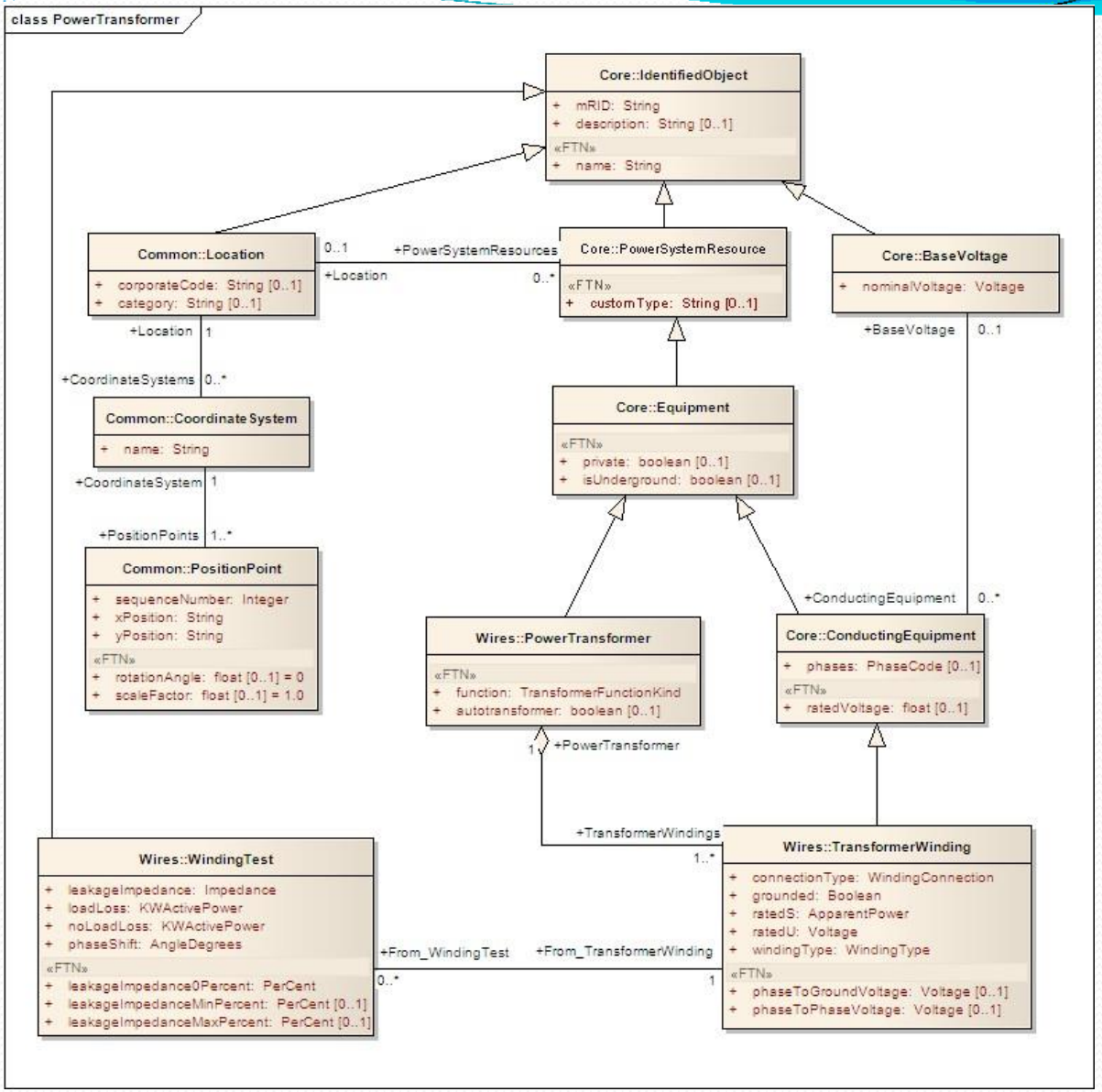
- Globalni identifikator
  - Globalni identifikator je 64 bitni
  - Kreira na osnovu tri podatka
    - 16 bitni *Sistem id* – za nas je uvek nula, jer koristimo jedan sistem
    - *DMSType* – odgovara tipu entiteta za koji se kreira globalni identifikator
    - *Brojač* – za svaki tip entitata postoji odgovarajući brojač koji obezbeđuje jedinstvenost globalnog identifikatora po tipu.



Reference – veze između entiteta



- Reference se modeluju globalnim identifikatorm
  - Za razliku od modela u *loader*-u koji je za referencu imao "pravu referencu" na drugi entitet, na servisu koristimo globalne identifikatore
  - Entitet koja referencira neki drugi entitet u stvari sadrži globalni identifikator tog entiteta
- Implementaciona specifičnost: Reference su dvosmerne na servisu: Ukoliko jedan entitet sadrži globalni identifikator drugog entiteta, onda i drugi entitet sadrži identifikator prvog.


class PowerTransformer
Core::IdentifiedObject
+ mRID: String
+ description: String [0..1]
«FTN»
+ name: String
Common::Location
+ corporateCode: String [0..1]
+ category: String [0..1]
0..1
+PowerSystemResources
+Location
0..*
Core::PowerSystemResource
«FTN»
+ customType: String [0..1]
Core::BaseVoltage
+ nominalVoltage: Voltage
+Location
1
+BaseVoltage
0..1
+CoordinateSystems
0..*
Common::CoordinateSystem
+ name: String
Core::Equipment
«FTN»
+ private: boolean [0..1]
+ isUnderground: boolean [0..1]
+CoordinateSystem
1
+PositionPoints
1..*
Common::PositionPoint
+ sequenceNumber: Integer
+ xPosition: String
+ yPosition: String
«FTN»
+ rotationAngle: float [0..1] = 0
+ scaleFactor: float [0..1] = 1.0
+ConductingEquipment
0..*
Wires::PowerTransformer
«FTN»
+ function: TransformerFunctionKind
+ autotransformer: boolean [0..1]
Core::ConductingEquipment
+ phases: PhaseCode [0..1]
«FTN»
+ ratedVoltage: float [0..1]
1
+PowerTransformer
+TransformerWindings
1..*
Wires::WindingTest
+ leakageImpedance: Impedance
+ loadLoss: KWActivePower
+ noLoadLoss: KWActivePower
+ phaseShift: AngleDegrees
«FTN»
+ leakageImpedance0Percent: PerCent
+ leakageImpedanceMinPercent: PerCent [0..1]
+ leakageImpedanceMaxPercent: PerCent [0..1]
+From_WindingTest
+From_TransformerWinding
0..*
1
Wires::TransformerWinding
+ connectionType: WindingConnection
+ grounded: Boolean
+ ratedS: ApparentPower
+ ratedU: Voltage
+ windingType: WindingType
«FTN»
+ phaseToGroundVoltage: Voltage [0..1]
+ phaseToPhaseVoltage: Voltage [0..1]

Zadaci

1. Kreirati *ModelCode*-ove za klase *TransformerWinding* i *WindingTest*